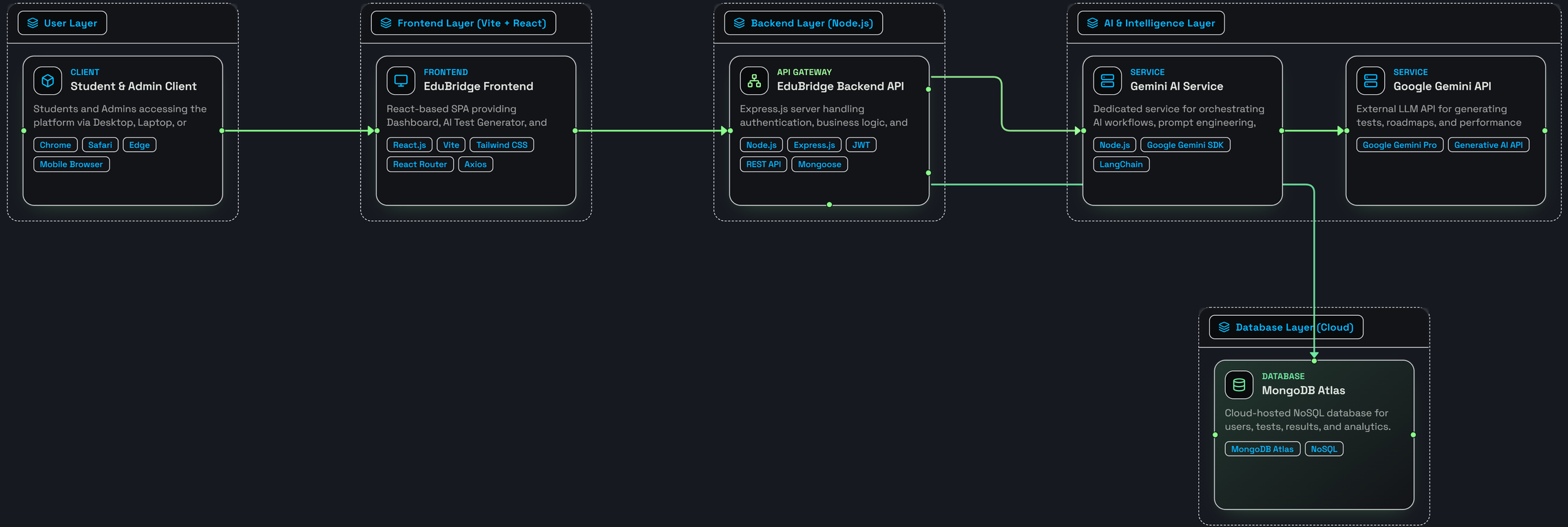
User Layer
CLIENT
Student & Admin Client
Students and Admins accessing the platform via Desktop, Laptop, or
Chrome
Safari
Edge
Mobile Browser
Frontend Layer (Vite + React)
FRONTEND
EduBridge Frontend
React-based SPA providing Dashboard, AI Test Generator, and
React.js
Vite
Tailwind CSS
React Router
Axios
Backend Layer (Node.js)
API GATEWAY
EduBridge Backend API
Express.js server handling authentication, business logic, and
Node.js
Express.js
JWT
REST API
Mongoose
AI & Intelligence Layer
SERVICE
Gemini AI Service
Dedicated service for orchestrating AI workflows, prompt engineering,
Node.js
Google Gemini SDK
LangChain
SERVICE
Google Gemini API
External LLM API for generating tests, roadmaps, and performance
Google Gemini Pro
Generative AI API
Database Layer (Cloud)
DATABASE
MongoDB Atlas
Cloud-hosted NoSQL database for users, tests, results, and analytics.
MongoDB Atlas
NoSQL